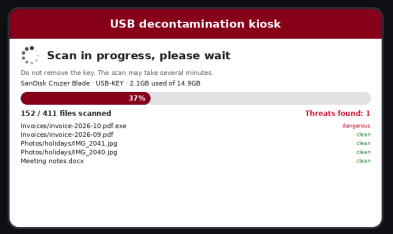


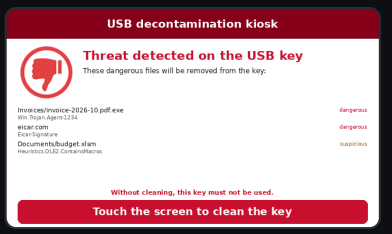
HACK.LU 2026 · LIVE DEMO

Pandora-Box

USB decontamination device

An open source, offline, hardened sheep-dip kiosk.
Plug in a USB key. Watch it get scanned. Touch to clean.





USB keys remain a common way into otherwise well-protected systems. **Pandora-Box** checks every file with several detection engines, shows the dangerous ones on a touch screen and removes them, **before the key reaches a workstation**. Free, auditable, hardened — for healthcare, industry and public administration.

01
Insert
Plug a USB key into the kiosk. No login, no buttons, no network.

02
Scan
Every file goes through ClamAV, YARA-X, MalwareBazaar, Hashlookup and heuristics.

03
Clean
Threats are listed, quarantined, then removed with one touch. A JSON report is kept.

- **Fail closed**
  A file that could not be fully scanned is never reported as clean.
- **Hardened image**
  Debian 13, read-only root with dm-verity, Secure Boot, USBGuard against BadUSB.
- **Sandboxed engines**
  Each engine runs in bubblewrap with no network access.
- **Signed updates**
  Signatures and A/B system updates, offline from USB, automatic rollback.
- **Cheap hardware**
  Refurbished Lenovo ThinkCentre Tiny + 7" Waveshare touch screen, 3D-printed case.
- **Customizable**
  Logo, colors, texts and language of the interface set by the organization.

STACK Python 3.11 · Pillow framebuffer UI (no X, no browser) · Debian 13 · mkosi · systemd-boot · systemd-sysupdate · dm-verity · USBGuard · bubblewrap · nftables · tested end-to-end in QEMU/KVM

SOURCE CODE

github.com/sourcentis/pandora-box

Docs: sourcentis.github.io/pandora-box · Contributions welcome!

GPLv3
Free software
Enclosure: CERN-OHL-S v2